

MarkPaul Chege Mureithi

Data Engineer | Software Developer

+254 748 280 758 | chegemark582@gmail.com | Nairobi, Kenya
github.com/chegeMP | linkedin.com/in/markchege4401 | chegeMP.github.io/Portfolio/

Summary

Data Engineer and Software Developer with hands-on experience designing and maintaining end-to-end data pipelines across distributed big-data infrastructure (NiFi, HDFS, Apache Iceberg, Trino) on Cloudera CDH. Comfortable across the full data lifecycle — data modeling, ETL/ELT, quality assurance, and automation — through to dashboarding. Experience collaborating with data science and business teams to turn requirements into production data assets. Certified in Python and Java, currently pursuing further certifications in Data Science, AI, and Machine Learning.

Education

Kabarak University
Bachelor of Computer Science
Sep 2021 — Dec 2025

Technical Skills

Python, SQL, Apache ecosystem (NiFi, Hadoop/HDFS, Iceberg, Trino), Java, Go, Power BI, Superset

Soft Skills

Collaboration, Problem-solving, Communication, Time management, Result-oriented

Additional Skills

Public Speaking, Writing, Research

Languages

English, Hindi

Certifications

- Certified Web Professional – Web Developer
- Java Development Certified Professional

Work Experience

Jamii Telecommunications Limited Data Engineer

Nov 2025 — Jul 2026

JTL is a leading internet service provider and telecommunications company in Kenya.

- Designed, built, and maintained end-to-end data pipelines (Python scraper → Apache NiFi → HDFS → Apache Iceberg → Trino → Superset) on Cloudera CDH infrastructure, replacing manual, ad-hoc reporting with a reliable, queryable analytics layer
- Designed data models and schemas for the analytics layer to support efficient querying across large, distributed datasets
- Led data validation and quality control checks across the pipeline, reducing the risk of flawed data reaching downstream analytics and machine learning models
- Collaborated with data science and business stakeholders to translate requirements into productionized data assets and self-serve reporting tools
- Built a Superset-style analytics interface from scratch, including a pivot table chart builder and dashboard layout system, giving non-technical stakeholders self-serve access to data
- Built JTL-Hub, an internal analytics platform (Go backend, Next.js/React frontend), consolidating dashboards and reporting into one internal tool
- Developed and evaluated machine learning models for churn prediction, extending the analytics layer into an early-warning signal for at-risk customers
- Documented technical workflows and data architecture, cutting onboarding time and reducing repeated troubleshooting for the team

Kocela Limited Software Developer Intern

Feb 2025 — May 2025

A startup focused on mobile money and financial technology.

- Built and maintained backend services and database logic (Play Framework, MySQL) supporting a mobile money/fintech platform
- Optimized database queries and backend integrations, improving data retrieval performance and application responsiveness
- Supported API integration work and bug fixes, improving data flow reliability between services

Nexora Technologies Data Science & Software Developer

Jan 2022 — Jul 2026

Independent contract work delivering data and software solutions for various clients.

- Built ETL-style workflows to clean, validate, and transform messy client datasets, making them usable for reporting and model training
- Delivered churn prediction and pricing models, and Power BI dashboards giving clients real-time visibility into KPIs, replacing manual reporting processes
- Provided ongoing technical support and pipeline optimization, keeping deployed solutions reliable after handoff

Khetias Supermarket Consultant

A chain of outlet supermarkets operating across the country.

- Resolved system and hardware/software issues across retail outlets, minimizing downtime for day-to-day operations
- Trained staff on business systems and digital tools, reducing dependency on IT support for routine tasks
- Supported IT infrastructure reliability through preventive maintenance, reducing recurring technical issues

Projects

ETL Pipeline for medical dataset

May 2026 — Jun 2026

github.com/chegeMP/ETL-1-medicine

- Built an ETL pipeline that ingests data and uses a cron job to clean and prepare it for visualization
- Eliminated manual handling of medical data, reducing risk of human error in a sensitive domain
- Cut turnaround time from raw medical data to reportable, visualization-ready output

Model Performance Dashboard

Nov 2024 — Jan 2025

github.com/chegeMP/Model-monitoring-dashboard

- Python dashboard for post-deployment model monitoring, tracking data drift (PSI, Wasserstein distance, mean shift) and performance stability (ROC-AUC) across baseline vs. comparison windows
- Enabled early detection of data drift and reduced risk of silent model failure by building automated tracking instead of manual, after-the-fact checks

Executive KPI Dashboard

Mar 2026

chegeMP.github.io/Portfolio/

- Power BI/SQL dashboard tracking revenue and product mix with drill-down views across business units, giving leadership month-over-month visibility without manual report compilation

Customer Churn Prediction Model

Jan 2026 — Mar 2026

chegeMP.github.io/Portfolio/

- Built a neural network churn prediction model (Python, TensorFlow/Keras) using feature engineering and class balancing; achieved Accuracy 0.8935, F1 0.8443, ROC-AUC 0.9670

Insurance Cost Predictor

Feb 2025

github.com/chegeMP/Insurance-Cost-Predictor1

- Machine learning pipeline (Flask, Python) for insurance pricing estimation using demographic and risk features, delivered through a web interface